Experiment-14

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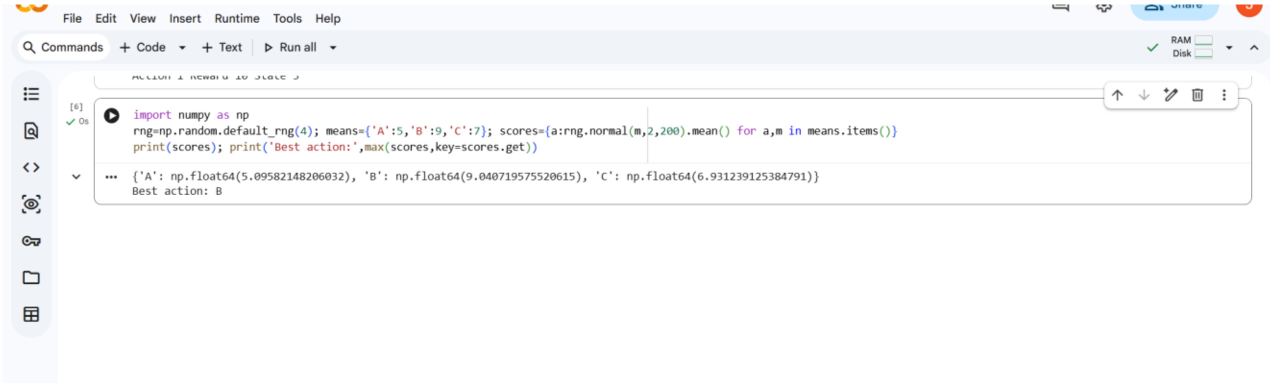
Course Code: -MLA0305 **Slot-**B

Title: Sampling-Based Planning Using a Learned Environment Model

Aim:
To implement sampling-based planning techniques for generating optimal action sequences using a learned environment model.

Procedure:

1. Develop or obtain an approximate environment model.
2. Observe the current state.
3. Generate multiple candidate action sequences.
4. Simulate each sequence using the learned model.
5. Calculate the expected cumulative reward for each sequence.
6. Select the sequence with the highest predicted reward.
7. Execute the first action of the selected sequence.
8. Observe the new state.
9. Repeat the planning process.
10. Compare predicted and actual rewards.



Result:

Sampling-based planning successfully generated action sequences using the learned model. The agent selected actions based on predicted future rewards.